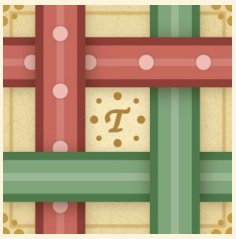
Repeating Knot Cards

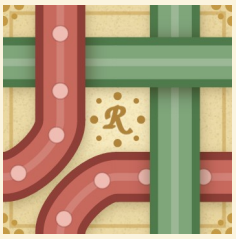
Justin Roughley, 2026

Four cards, nine wallpaper groups

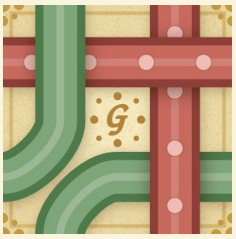
A pack of 60: forty-eight woven cards, nine target cards and three reference cards. Every card is a little patch of Celtic knotwork, and every one can be turned any way: wherever it goes, the ribbons still pass alternately over and under. Lay four cards in a square, repeat the square, and the knot takes on one of nine kinds of symmetry: nine of the seventeen wallpaper groups (en.wikipedia.org/wiki/Wallpaper_group).



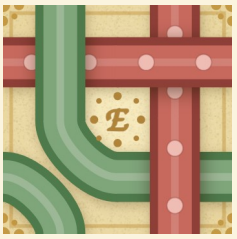
T × 16
four crossings: a trellis



R × 8
two red bends



G × 16
two green bends



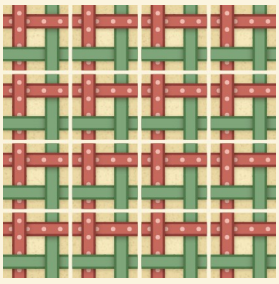
E × 8
an elbow and a corner

FIND THE GROUP

1. Pick a target card. Lay out the four cards on its front, each turned so its letter faces as shown.
2. Make three more squares the same and join them: sixteen cards. Colours must match across every edge.
3. Look for mirrors, glides and turns, and name the group. Over and under count; colours do not.
4. Turn the target over to check: it names the group and marks its mirrors, glides and turns.

THE NINE TARGETS

With colours gives the group when the colours count too; *Shape only*, when over and under are ignored.

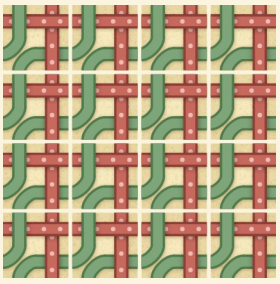


1 **4*2** (p4g)

T · T
T · T

The plain weave: quarter turns and mirrors

With colours ***2222**
Shape only ***442**

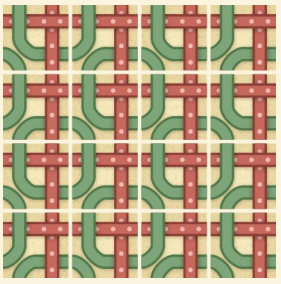


2 **2222** (p2)

G · G
G · G

Half turns only

With colours same
Shape only **2*22**

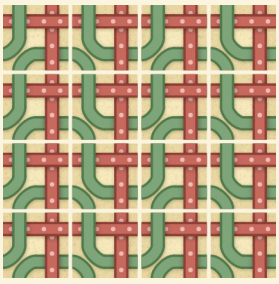


3 **2*22** (cmm)

E · G
G · E

Mirrors in two directions, half turns

With colours same
Shape only ***442**

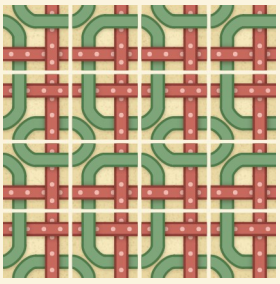


4 **22*** (pmg)

G · E
G · E

Parallel mirrors, half turns, glides

With colours same
Shape only same

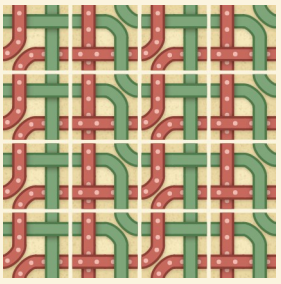


5 ****** (pm)

G ↻1 · E ↻1
G · E

Parallel mirrors only

With colours same
Shape only ***2222**

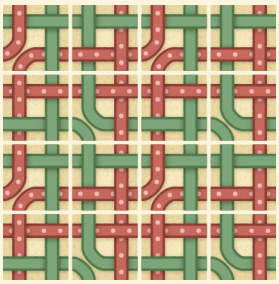


6 **××** (pg)

R · E ↻2
R · E ↻2

Glides only: reflect and slide

With colours **o**
Shape only same

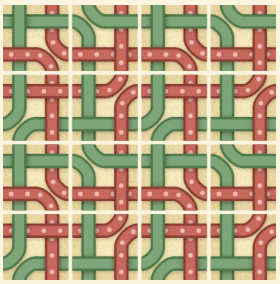


7 **22×** (pgg)

R · T ↻2
T · E

Half turns and glides, no mirrors

With colours **o**
Shape only **4*2**

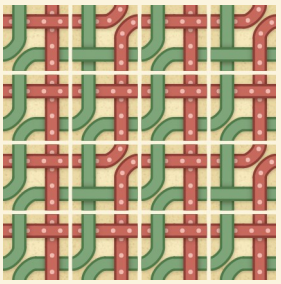


8 **442** (p4)

R ↻3 · G ↻1
G · R ↻2

Quarter turns, no mirrors

With colours **o**
Shape only **4*2**



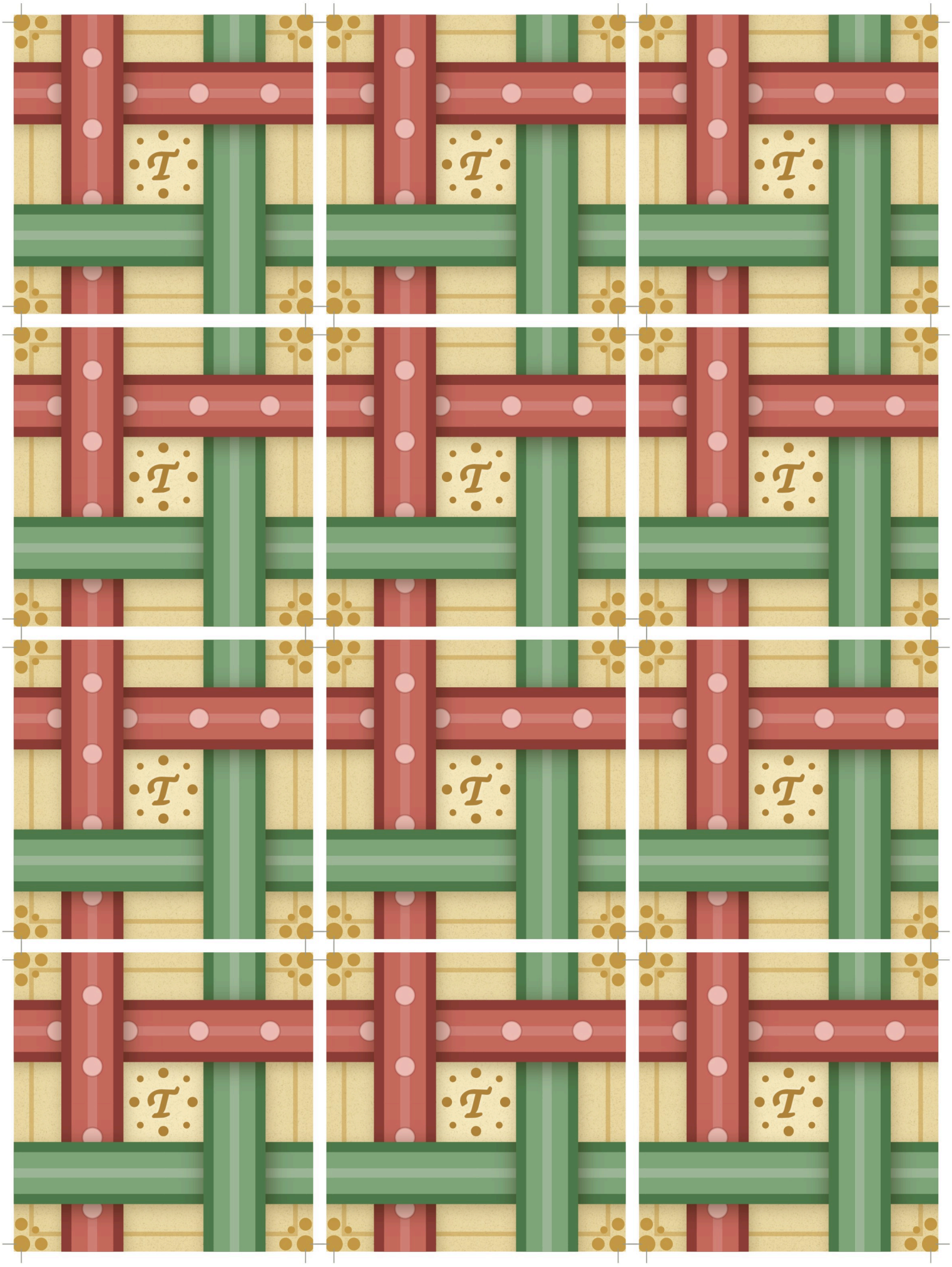
9 **o** (p1)

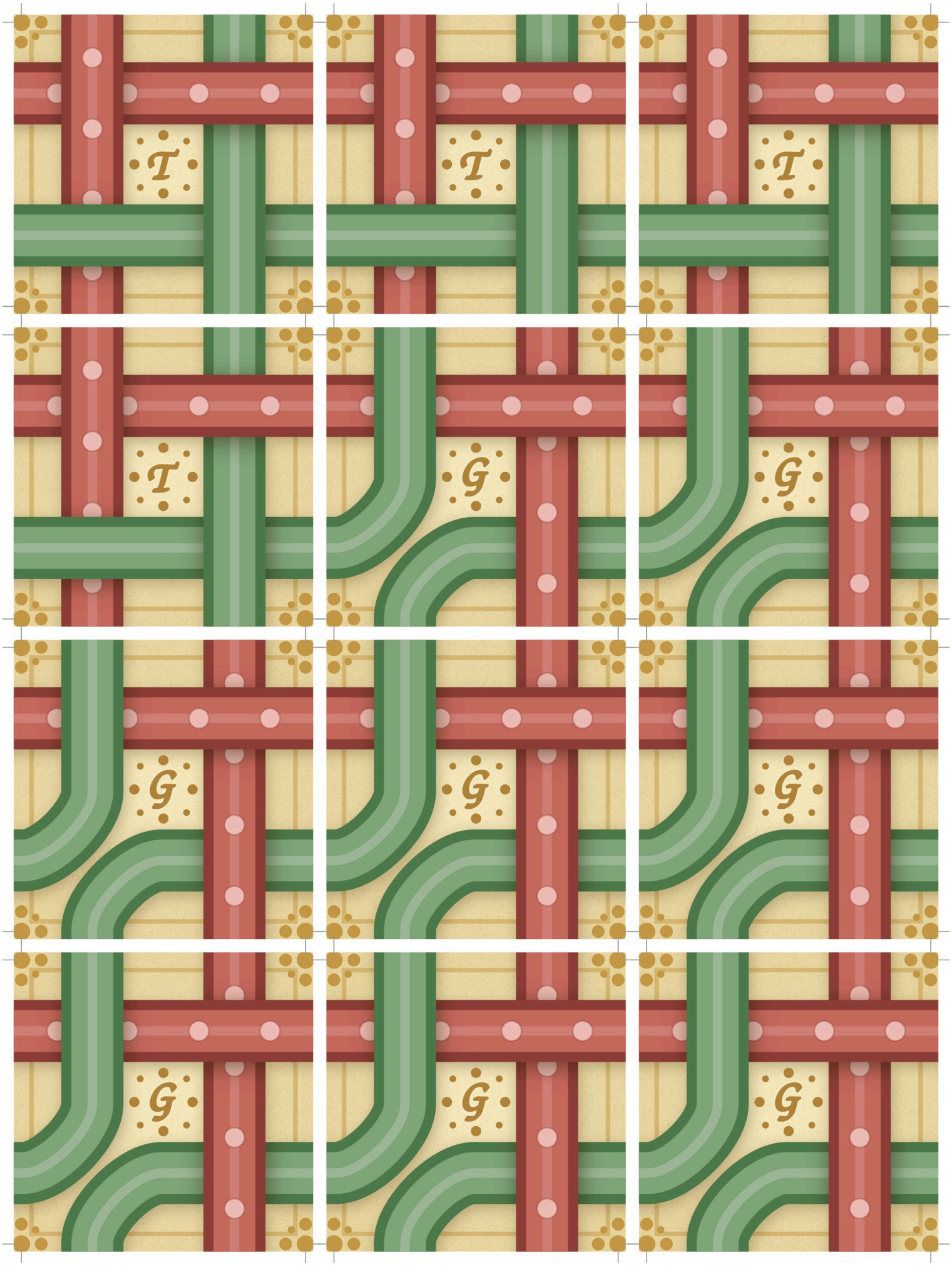
G · R ↻2
G · G

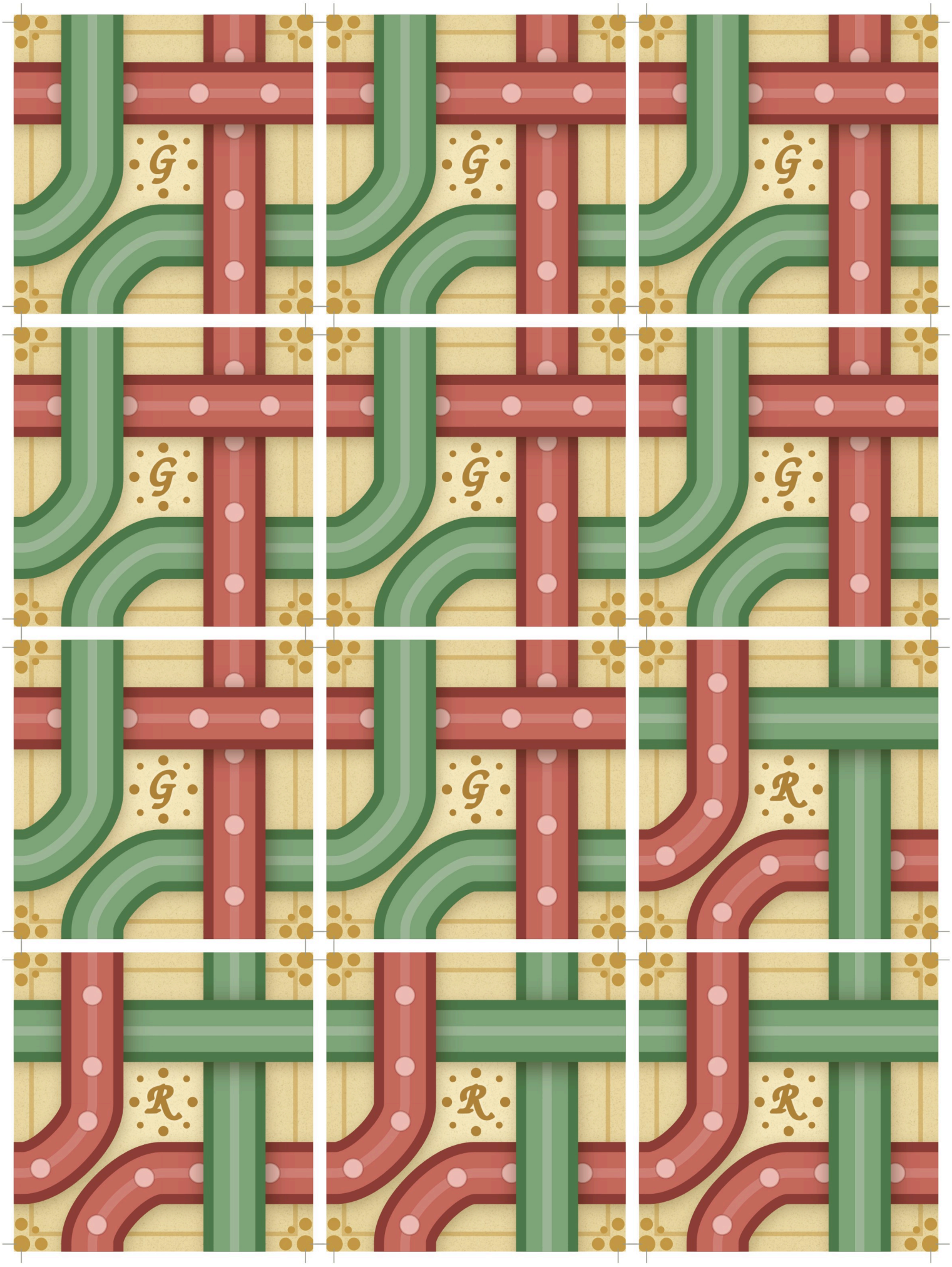
Translations only

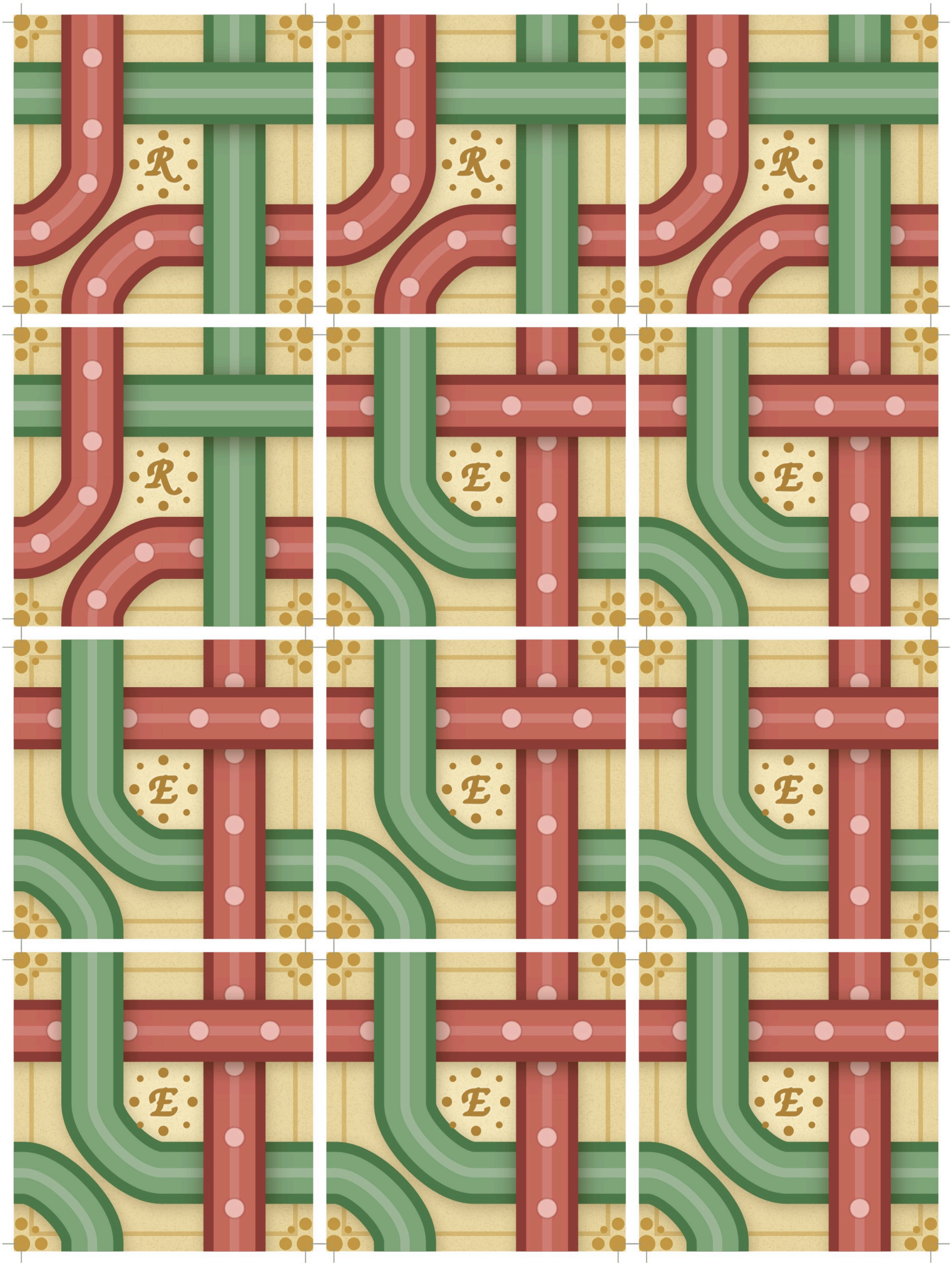
With colours same
Shape only ******

Free to print from https://github.com/Cortado-J/REPEATING_KNOT/tree/main/KNOT_CARDS




T
T
T
T
G
G
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G


G
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R
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E

Repeating Knot Cards

four cards · nine wallpaper groups

1. Pick a target card, recipe side up.
2. Lay out its four cards in a square, each turned so its letter faces as shown.
3. Make four such squares and join them: sixteen cards. Match ribbon colours across every edge.
4. Find the mirrors, glides and turns. Which group is it?
5. Turn the target over to check.

READING THE BACKS

- mirror
- glide: reflect and slide
- half-turn centre
- quarter-turn centre

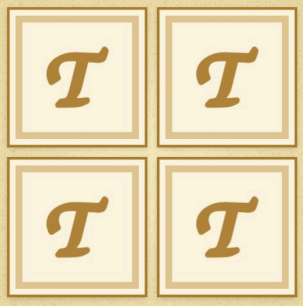
The name lists what a pattern has: a number for each kind of turn centre; ***** for mirrors, with a number after it for each kind of corner where mirrors meet; **×** for a glide that crosses no mirror; **o** for repeats only. In brackets: the crystallographers' name.

TO WONDER ABOUT

1. Why can't these cards make ***×**, the tenth group a square grid allows?
2. Change one card at a time. Can you visit all nine groups without repeating one?
3. How many strands of each colour does each pattern have?

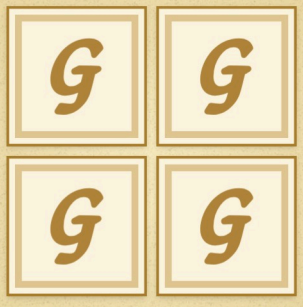


TARGET 1



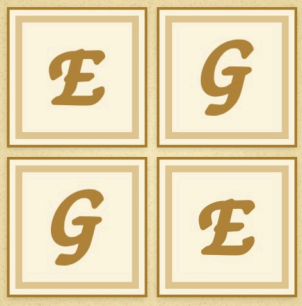
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 2



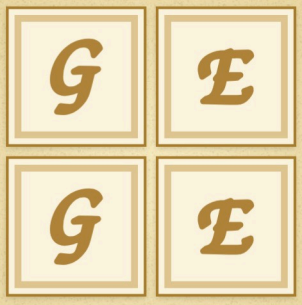
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 3



Make this square, then four of them. Which group is it? Turn over to check.

TARGET 4



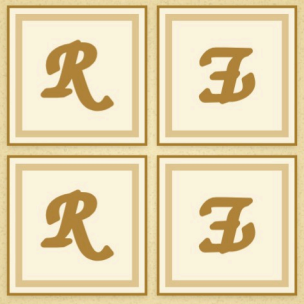
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 5



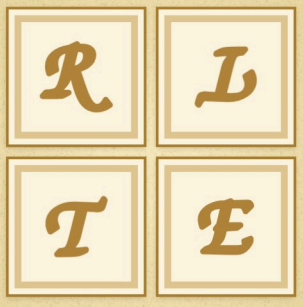
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 6



Make this square, then four of them. Which group is it? Turn over to check.

TARGET 7



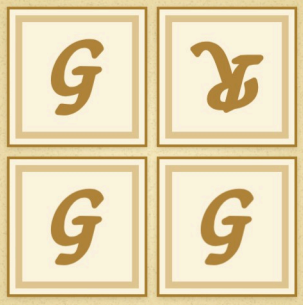
Make this square, then four of them. Which group is it? Turn over to check.

TARGET 8

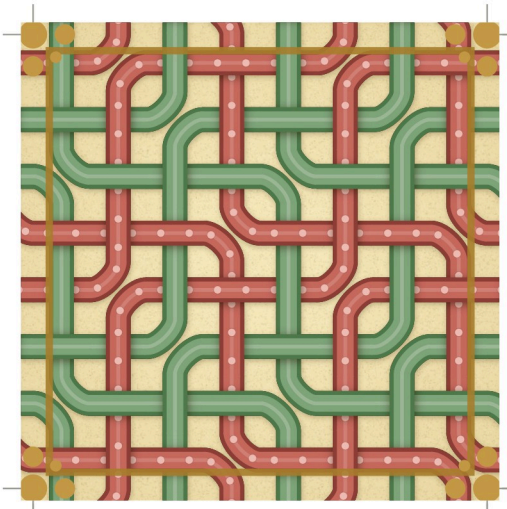


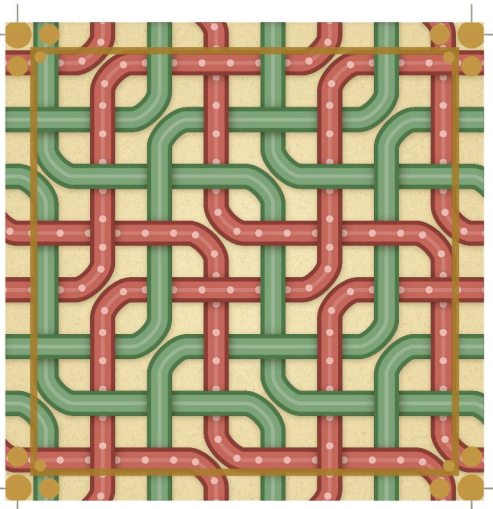
Make this square, then four of them. Which group is it? Turn over to check.

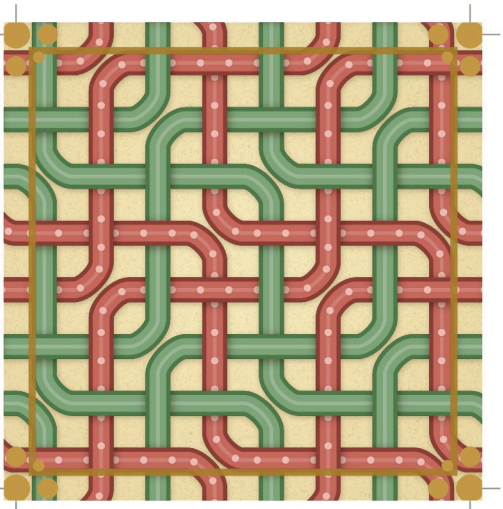
TARGET 9



Make this square, then four of them. Which group is it? Turn over to check.



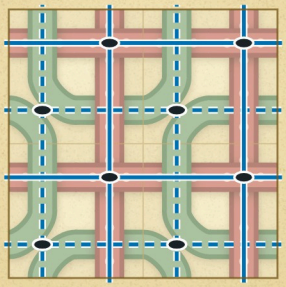




TARGET 3

2*22 (cmm)

Mirrors in two directions, half turns

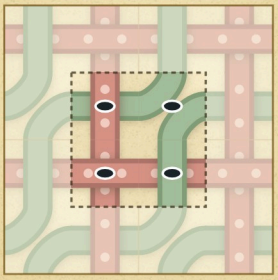


With colours same · Shape only ***442**

TARGET 2

2222 (p2)

Half turns only

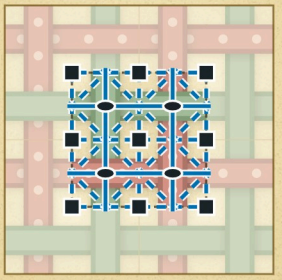


With colours same · Shape only **2*22**

TARGET 1

4*2 (p4g)

The plain weave: quarter turns and mirrors

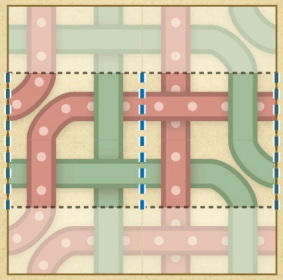


With colours ***2222** · Shape only ***442**

TARGET 6

×× (pg)

Glides only: reflect and slide

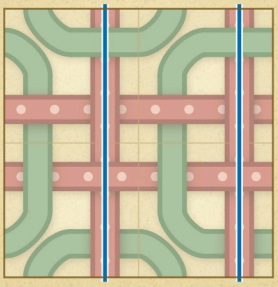


With colours **o** · Shape only same

TARGET 5

****** (pm)

Parallel mirrors only

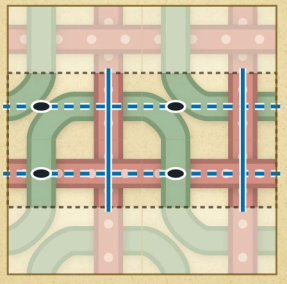


With colours same · Shape only ***2222**

TARGET 4

22* (pmg)

Parallel mirrors, half turns, glides

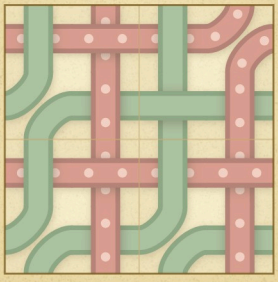


With colours same · Shape only same

TARGET 9

o (p1)

Translations only

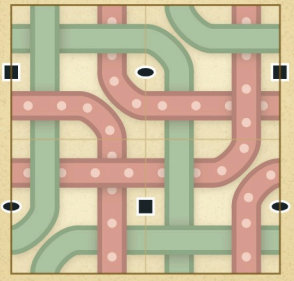


With colours same · Shape only ****

TARGET 8

442 (p4)

Quarter turns, no mirrors

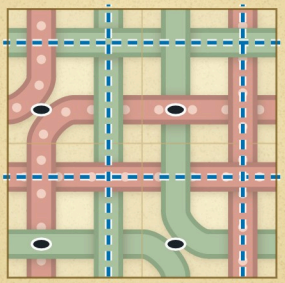


With colours **o** · Shape only **4*2**

TARGET 7

22× (pgg)

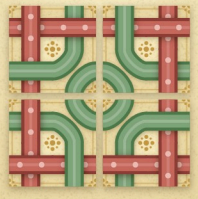
Half turns and glides, no mirrors



With colours **o** · Shape only **4*2**

Repeating Knot Cards

four cards · nine wallpaper groups



A PACK OF 60

Repeating Knot Cards

https://github.com/Cortado-J/REPEATING_KNOT/tree/main/KNOT_CARDS

Justin Roughley